PRAKTIKUM BASIS DATA

MODUL 5

Scripting Agregate Function

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

TUGAS PENDAHULUAN

1. Kerjakan contoh kasus pada modul 5 (halaman 61). Sertakan Sreenshoot untuk setiap nomor.
2. Buatlah table perbedaan antara Agregate Function dan Scalar Function. Berikan penjelasan dan contoh nya.

Jawaban:

Oracle SQL Developer : TP_Modul05_Basdat_103122430009

File Edit View Navigate Run Source Team Tools Window Help

TP_Modul05_Basdat_103122430009.sql

Worksheet | Query Builder

```
-- ----------------------------
-- Foreign Keys structure for table INVENTORI
-- ----------------------------
ALTER TABLE "INVENTORI" ADD CONSTRAINT "INVENTORI_FK1" FOREIGN KEY ("ID_THEATER") REFERENCES "THEATER" ("ID_THEATER") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;

-- ----------------------------
-- Foreign Keys structure for table JADWAL
-- ----------------------------
ALTER TABLE "JADWAL" ADD CONSTRAINT "JADWAL_FK1" FOREIGN KEY ("ID_FILM") REFERENCES "FILM" ("ID_FILM") ON DELETE CASCADE NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;
ALTER TABLE "JADWAL" ADD CONSTRAINT "JADWAL_FK2" FOREIGN KEY ("ID_THEATER") REFERENCES "THEATER" ("ID_THEATER") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;

-- ----------------------------
-- Foreign Keys structure for table TRANSAKSI
-- ----------------------------
ALTER TABLE "TRANSAKSI" ADD CONSTRAINT "TRANSAKSI_FK1" FOREIGN KEY ("ID_JADWAL") REFERENCES "JADWAL" ("ID_JADWAL") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;
ALTER TABLE "TRANSAKSI" ADD CONSTRAINT "TRANSAKSI_FK2" FOREIGN KEY ("ID_INVENTORI") REFERENCES "INVENTORI" ("ID_INVENTORI") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;
ALTER TABLE "TRANSAKSI" ADD CONSTRAINT "TRANSAKSI_FK3" FOREIGN KEY ("ID_MEMBER") REFERENCES "MEMBER" ("ID_MEMBER") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;

select count (*) AS JUMLAH_FILM_TERSEDIA
from FILM;

select count(DISTINCT ID_FILM) AS JUMLAH_FILM_DIPUTAR
```

Script Output | Query Result

All Rows Fetched: 1 in 0.004 seconds

	JUMLAH_FILM_TERSEDIA
1	4

Saved: D:\TUGAS RPL\Tugas Kuliah Semester 4\Basis Data\Praktikum\BASDAT_103122430009_Izzan-Maula-Rifqi_SE-08-02\05_Scripting_Agregate_Function\TP\TP_Modul05_Basdat_103122430009.sql

Oracle SQL Developer : TP_Modul05_Basdat_103122430009

File Edit View Navigate Run Source Team Tools Window Help

TP_Modul05_Basdat_103122430009.sql

Worksheet | Query Builder

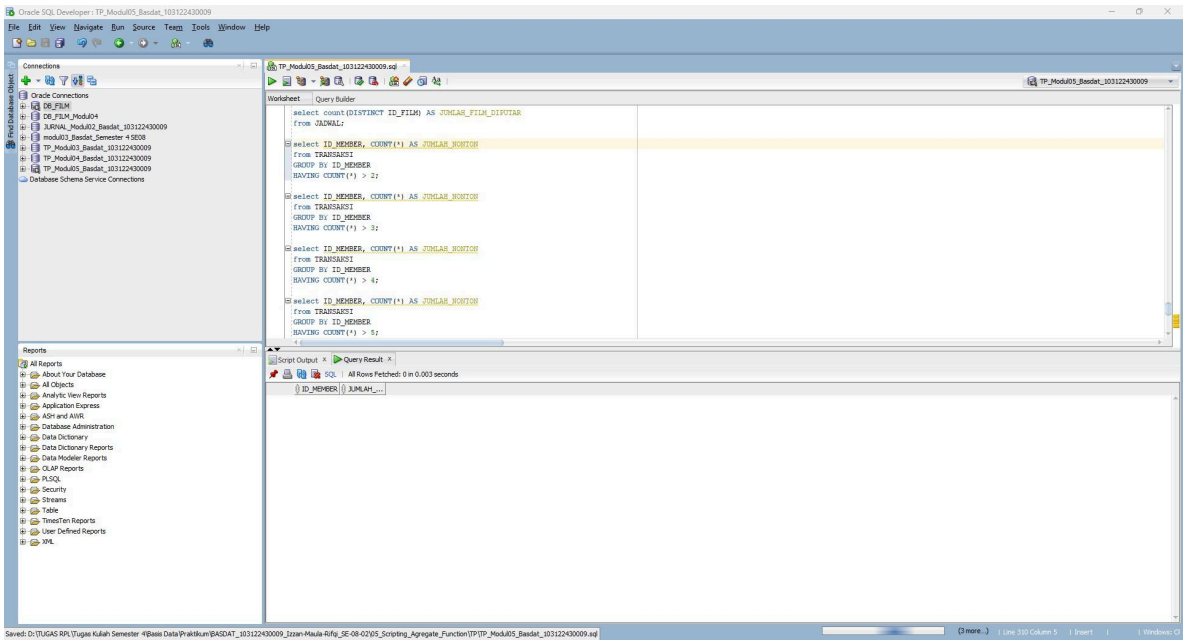
```
ALTER TABLE "INVENTORI" ADD CONSTRAINT "INVENTORI_FK1" FOREIGN KEY ("ID_THEATER") REFERENCES "THEATER" ("ID_THEATER") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;

-- ----------------------------
-- Foreign Keys structure for table JADWAL
-- ----------------------------
ALTER TABLE "JADWAL" ADD CONSTRAINT "JADWAL_FK1" FOREIGN KEY ("ID_FILM") REFERENCES "FILM" ("ID_FILM") ON DELETE CASCADE NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;
ALTER TABLE "JADWAL" ADD CONSTRAINT "JADWAL_FK2" FOREIGN KEY ("ID_THEATER") REFERENCES "THEATER" ("ID_THEATER") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;

-- ----------------------------
-- Foreign Keys structure for table TRANSAKSI
-- ----------------------------
ALTER TABLE "TRANSAKSI" ADD CONSTRAINT "TRANSAKSI_FK1" FOREIGN KEY ("ID_JADWAL") REFERENCES "JADWAL" ("ID_JADWAL") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;
ALTER TABLE "TRANSAKSI" ADD CONSTRAINT "TRANSAKSI_FK2" FOREIGN KEY ("ID_INVENTORI") REFERENCES "INVENTORI" ("ID_INVENTORI") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;
ALTER TABLE "TRANSAKSI" ADD CONSTRAINT "TRANSAKSI_FK3" FOREIGN KEY ("ID_MEMBER") REFERENCES "MEMBER" ("ID_MEMBER") NOT DEFERRABLE INITIALLY IMMEDIATE NORELY VALIDATE;

select count (*) AS JUMLAH_FILM_TERSEDIA
from FILM;

select count(DISTINCT ID_FILM) AS JUMLAH_FILM_DIPUTAR
from JADWAL;

select ID_MEMBER, COUNT(*) AS JUMLAH_NONTON
```

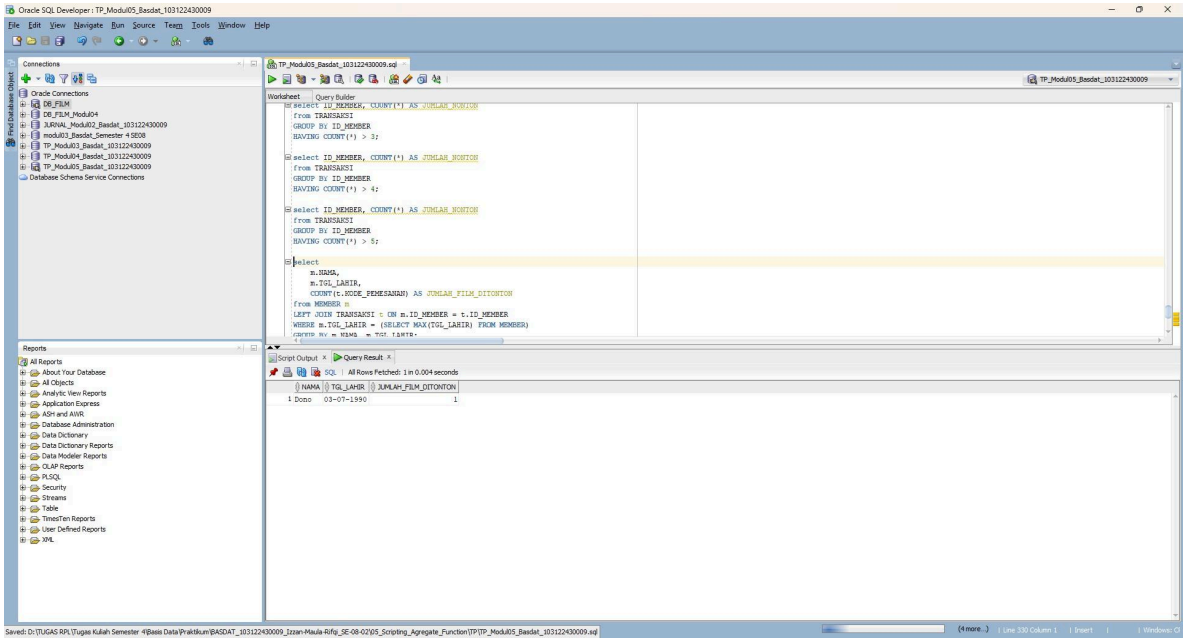
Script Output | Query Result

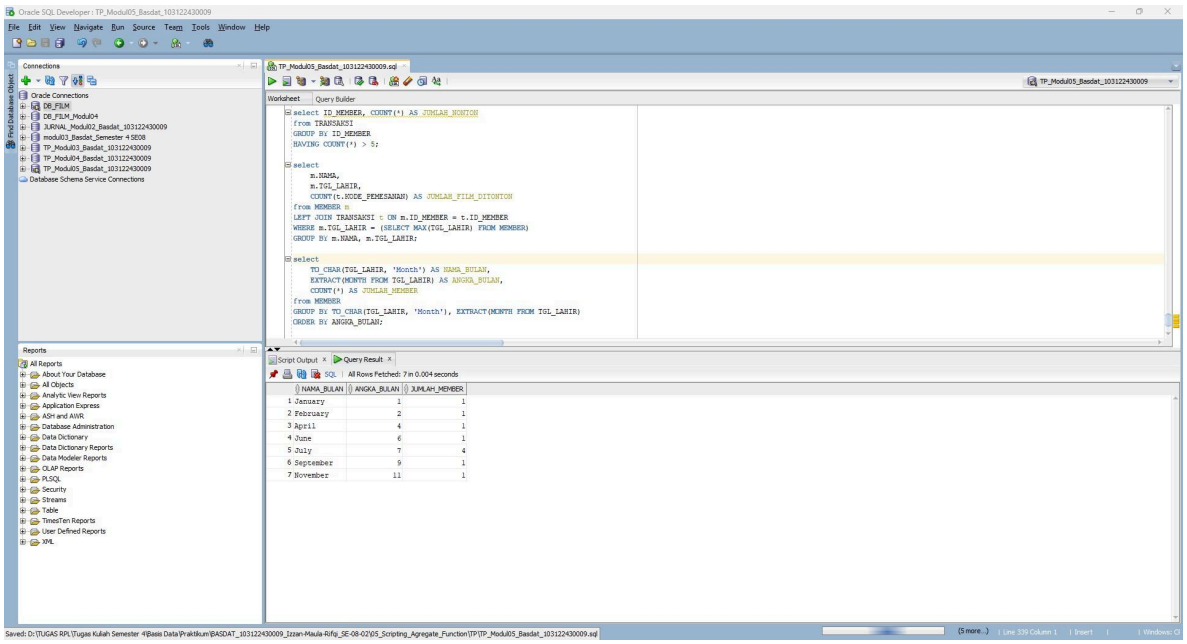
All Rows Fetched: 1 in 0.003 seconds

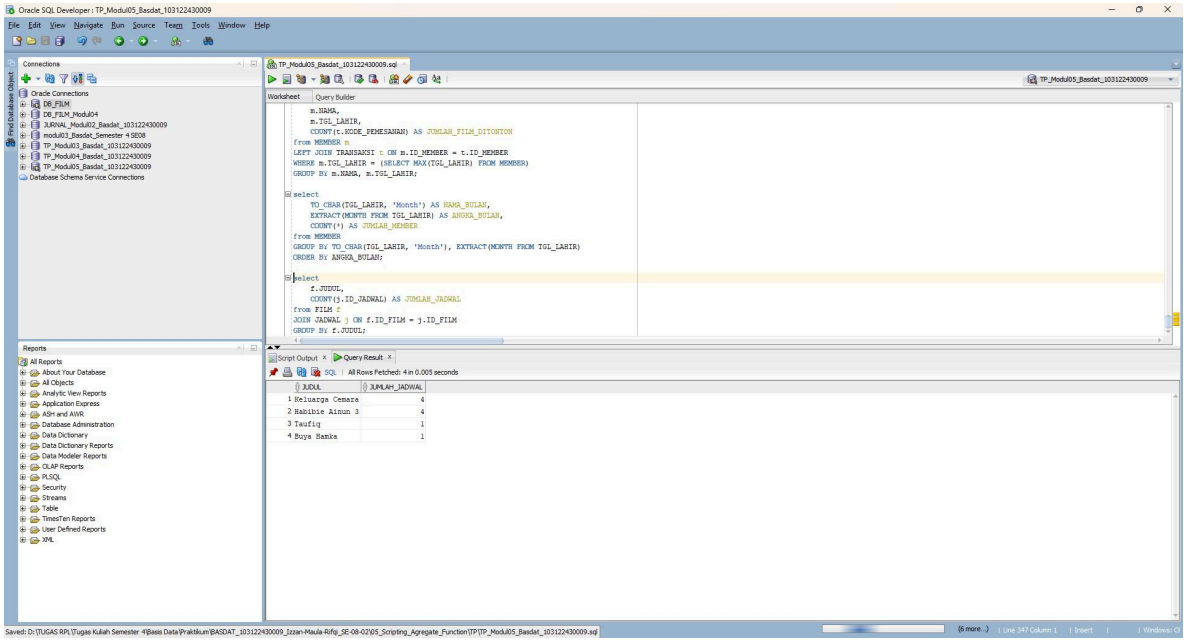
	JUMLAH_FILM_DIPUTAR
1	4

Saved: D:\TUGAS RPL\Tugas Kuliah Semester 4\Basis Data\Praktikum\BASDAT_103122430009_Izzan-Maula-Rifqi_SE-08-02\05_Scripting_Agregate_Function\TP\TP_Modul05_Basdat_103122430009.sql


Oracle SQL Developer : TP_Modul05_Basdat_103122430009
File Edit View Navigate Run Source Team Tools Window Help
Connections
Oracle Connections
DB_FILM
DB_FILM_Modul04
JURNAL_Modul02_Basdat_103122430009
modul03_Basdat_Semester 4 SE08
TP_Modul03_Basdat_103122430009
TP_Modul04_Basdat_103122430009
TP_Modul05_Basdat_103122430009
Database Schema Service Connections
TP_Modul05_Basdat_103122430009.sql
Worksheet
Query Builder
select count(DISTINCT ID_FILM) AS JUMLAH_FILM_DIPUTAR
from JADWAL;
select ID_MEMBER, COUNT(*) AS JUMLAH_NONTON
from TRANSAKSI
GROUP BY ID_MEMBER
HAVING COUNT(*) > 2;
select ID_MEMBER, COUNT(*) AS JUMLAH_NONTON
from TRANSAKSI
GROUP BY ID_MEMBER
HAVING COUNT(*) > 3;
select ID_MEMBER, COUNT(*) AS JUMLAH_NONTON
from TRANSAKSI
GROUP BY ID_MEMBER
HAVING COUNT(*) > 4;
select ID_MEMBER, COUNT(*) AS JUMLAH_NONTON
from TRANSAKSI
GROUP BY ID_MEMBER
HAVING COUNT(*) > 5;
Reports
All Reports
About Your Database
All Objects
Analytic View Reports
Application Express
ASH and AWR
Database Administration
Data Dictionary
Data Dictionary Reports
Data Modeler Reports
OLAP Reports
PLSQL
Security
Streams
Table
TimesTen Reports
User Defined Reports
XML
Script Output
Query Result
All Rows Fetched: 0 in 0.003 seconds
ID_MEMBER
JUMLAH_...
Saved: D:\TUGAS RPL\Tugas Kuliah Semester 4\Basis Data\Praktikum\BASDAT_103122430009_Izzan-Maula-Rifqi_SE-08-02\05_Scripting_Agregate_Function\TP\TP_Modul05_Basdat_103122430009.sql


Oracle SQL Developer : TP_Modul05_Basdat_103122430009
File Edit View Navigate Run Source Team Tools Window Help
Connections
Oracle Connections
DB_FILM
DB_FILM_Modul04
JURNAL_Modul02_Basdat_103122430009
modul03_Basdat_Semester 4 SE08
TP_Modul03_Basdat_103122430009
TP_Modul04_Basdat_103122430009
TP_Modul05_Basdat_103122430009
Database Schema Service Connections
TP_Modul05_Basdat_103122430009.sql
Worksheet
Query Builder
select ID_MEMBER, COUNT(*) AS JUMLAH_NONTON
from TRANSAKSI
GROUP BY ID_MEMBER
HAVING COUNT(*) > 3;
select ID_MEMBER, COUNT(*) AS JUMLAH_NONTON
from TRANSAKSI
GROUP BY ID_MEMBER
HAVING COUNT(*) > 4;
select ID_MEMBER, COUNT(*) AS JUMLAH_NONTON
from TRANSAKSI
GROUP BY ID_MEMBER
HAVING COUNT(*) > 5;
select
m.NAMA,
m.TGL_LAHIR,
COUNT(t.KODE_PEMESANAN) AS JUMLAH_FILM_DITONTON
from MEMBER m
LEFT JOIN TRANSAKSI t ON m.ID_MEMBER = t.ID_MEMBER
WHERE m.TGL_LAHIR = (SELECT MAX(TGL_LAHIR) FROM MEMBER)
Reports
All Reports
About Your Database
All Objects
Analytic View Reports
Application Express
ASH and AWR
Database Administration
Data Dictionary
Data Dictionary Reports
Data Modeler Reports
OLAP Reports
PLSQL
Security
Streams
Table
TimesTen Reports
User Defined Reports
XML
Script Output
Query Result
All Rows Fetched: 1 in 0.004 seconds
NAMA
TGL_LAHIR
JUMLAH_FILM_DITONTON
1 Dono 03-07-1990 1
Saved: D:\TUGAS RPL\Tugas Kuliah Semester 4\Basis Data\Praktikum\BASDAT_103122430009_Izzan-Maula-Rifqi_SE-08-02\05_Scripting_Agregate_Function\TP\TP_Modul05_Basdat_103122430009.sql


Oracle SQL Developer : TP_ModulI05_Basdat_103122430009
File Edit View Navigate Run Source Team Tools Window Help
Connections
Oracle Connections
DB_FILM
DB_FILM_ModulI04
JURNAL_ModulI02_Basdat_103122430009
modulI03_Basdat_Semester 4 SE08
TP_ModulI03_Basdat_103122430009
TP_ModulI04_Basdat_103122430009
TP_ModulI05_Basdat_103122430009
Database Schema Service Connections
TP_ModulI05_Basdat_103122430009.sql
Worksheet Query Builder
select ID_MEMBER, COUNT(*) AS JUMLAH_NONTON
from TRANSAKSI
GROUP BY ID_MEMBER
HAVING COUNT(*) > 5;
select
m.NAMA,
m.TGL_LAHIR,
COUNT(t.KODE_PEMESANAN) AS JUMLAH_FILM_DITONTON
from MEMBER m
LEFT JOIN TRANSAKSI t ON m.ID_MEMBER = t.ID_MEMBER
WHERE m.TGL_LAHIR = (SELECT MAX(TGL_LAHIR) FROM MEMBER)
GROUP BY m.NAMA, m.TGL_LAHIR;
select
TO_CHAR(TGL_LAHIR, 'Month') AS NAMA_BULAN,
EXTRACT(MONTH FROM TGL_LAHIR) AS ANGKA_BULAN,
COUNT(*) AS JUMLAH_MEMBER
from MEMBER
GROUP BY TO_CHAR(TGL_LAHIR, 'Month'), EXTRACT(MONTH FROM TGL_LAHIR)
ORDER BY ANGKA_BULAN;
Reports
All Reports
About Your Database
All Objects
Analytic View Reports
Application Express
ASH and AWR
Database Administration
Data Dictionary
Data Dictionary Reports
Data Modeler Reports
OLAP Reports
PLSQL
Security
Streams
Table
TimesTen Reports
User Defined Reports
XML
Script Output
Query Result
All Rows Fetched: 7 in 0.004 seconds
NAMA_BULAN ANGKA_BULAN JUMLAH_MEMBER
1 January 1 1
2 February 2 1
3 April 4 1
4 June 6 1
5 July 7 4
6 September 9 1
7 November 11 1
Saved: D:\TUGAS RPL\Tugas Kuliah Semester 4\Basis Data\Praktikum\BASDAT_103122430009_Izzan-Maula-Rifqi_SE-08-02\05_Scripting_Agregate_Function\TP\TP_ModulI05_Basdat_103122430009.sql
(5 more...) | Line 339 Column 1 | Insert


Oracle SQL Developer : TP_ModulI05_Basdat_103122430009
File Edit View Navigate Run Source Team Tools Window Help
Connections
Oracle Connections
DB_FILM
DB_FILM_ModulI04
JURNAL_ModulI02_Basdat_103122430009
modulI03_Basdat_Semester 4 SE08
TP_ModulI03_Basdat_103122430009
TP_ModulI04_Basdat_103122430009
TP_ModulI05_Basdat_103122430009
Database Schema Service Connections
TP_ModulI05_Basdat_103122430009.sql
Worksheet Query Builder
m.NAMA,
m.TGL_LAHIR,
COUNT(t.KODE_PEMESANAN) AS JUMLAH_FILM_DITONTON
from MEMBER m
LEFT JOIN TRANSAKSI t ON m.ID_MEMBER = t.ID_MEMBER
WHERE m.TGL_LAHIR = (SELECT MAX(TGL_LAHIR) FROM MEMBER)
GROUP BY m.NAMA, m.TGL_LAHIR;
select
TO_CHAR(TGL_LAHIR, 'Month') AS NAMA_BULAN,
EXTRACT(MONTH FROM TGL_LAHIR) AS ANGKA_BULAN,
COUNT(*) AS JUMLAH_MEMBER
from MEMBER
GROUP BY TO_CHAR(TGL_LAHIR, 'Month'), EXTRACT(MONTH FROM TGL_LAHIR)
ORDER BY ANGKA_BULAN;
select
f.JUDUL,
COUNT(j.ID_JADWAL) AS JUMLAH_JADWAL
from FILM f
JOIN JADWAL j ON f.ID_FILM = j.ID_FILM
GROUP BY f.JUDUL;
Reports
All Reports
About Your Database
All Objects
Analytic View Reports
Application Express
ASH and AWR
Database Administration
Data Dictionary
Data Dictionary Reports
Data Modeler Reports
OLAP Reports
PLSQL
Security
Streams
Table
TimesTen Reports
User Defined Reports
XML
Script Output
Query Result
All Rows Fetched: 4 in 0.005 seconds
JUDUL JUMLAH_JADWAL
1 Keluarga Cemara 4
2 Habibie Ainun 3 4
3 Taufiq 1
4 Buya Hamka 1
Saved: D:\TUGAS RPL\Tugas Kuliah Semester 4\Basis Data\Praktikum\BASDAT_103122430009_Izzan-Maula-Rifqi_SE-08-02\05_Scripting_Agregate_Function\TP\TP_ModulI05_Basdat_103122430009.sql
(6 more...) | Line 347 Column 1 | Insert

2. Table perbedaan antara Agregate Function dan Scalar Function

No.	Perbandingan	Aggregate Function	Scalar Function
1.	Pengertian	Fungsi yang memproses banyak baris data secara bersamaan untuk menghasilkan satu nilai ringkasan.	Fungsi yang memproses satu baris data tunggal dan menghasilkan satu nilai baru untuk setiap baris tersebut.
2.	Pola Input - Output	Many-to-One	One-to-One
3.	Tujuan Penggunaan	Untuk menghitung total, rata-rata, jumlah, atau mencari nilai tertinggi/terendah dari sebuah kelompok data.	Untuk memanipulasi teks seperti besar atau kecilnya huruf, menghitung panjang huruf, membulatkan angka, atau mengubah format tanggal.
4.	Contoh Fungsi	SUM(), COUNT(), AVG(), MAX(), MIN()	UPPER(), LOWER(), LENGTH(), ROUND(), SUBSTR()
5.	Penggunaan GROUP BY	Sangat sering digunakan bersama GROUP BY untuk meringkas data per kategori.	Tidak membutuhkan GROUP BY.